

# Producing Benzoic Acid (Cambridge (CIE) A Level Chemistry): Revision Note

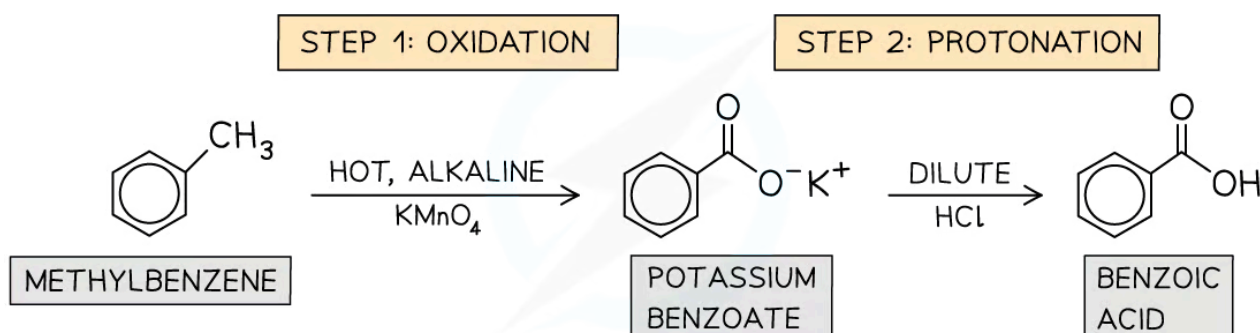
## Production of Benzoic Acid

- **Benzoic acids** are the simplest **aromatic carboxylic acids** with the molecular formula of  $\text{C}_6\text{H}_5\text{COOH}$
- Benzoic acids and their derivatives are often used as reagents in the synthesis of **esters**
- The compounds can be produced from the **oxidation** of **alkylbenzenes**

## Oxidation of alkylbenzenes

- The alkyl side-chain in alkylbenzenes, such as methylbenzene, can be oxidised to a **carboxylic acid**
- The alkylbenzene is **heated under reflux** with a solution of **hot alkaline  $\text{KMnO}_4$**  (this is the **oxidising agent**)
  - The **purple** colour of the  $\text{Mn}^{7+}$  ions disappears as they are **reduced** to  $\text{Mn}^{4+}$  ions
  - A **brown precipitate** of  $\text{MnO}_2$  is formed
- The mixture is then **acidified** with **dilute acid** (such as hydrochloric acid) to **protonate** the organic product form and produce a benzoic acid

## Making benzoic acid from methylbenzene



*Alkylbenzenes such as methylbenzene undergo oxidation to form benzoic acid*